

1. [5 pts] Find the general antiderivative of the following functions.

1). $f(x) = 4 \sin x$

Note: $F(x)$ is the general antiderivative.

$$F(x) = -4 \cos x + C$$

2). [3 pts] $f(x) = \frac{1}{x^4} = x^{-4}$

$$F(x) = -\frac{1}{3} x^{-3} + C$$

2. [5 pts] Find the following indefinite integrals.

1). $\int \frac{1}{1+x^2} dx = \arctan x + C$

2). $\int \frac{x^4 + x^2}{x^3} dx = \int (x + \frac{1}{x}) dx = \frac{1}{2} x^2 + \ln|x| + C.$